INCH-POUND

MIL-PRF-1/30B
22 January 2007
SUPERSEDING
MIL-E-1/30A
13 February 1969

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE, NEGATIVE GRID

TYPE 2C36

Inactive for new design
after 7 March 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification sheet and MIL-PRF-1.

DESCRIPTION: Triode, planar.

See figure 1.

Mounting position: Any.

Weight: 0.4 ounce nominal.

ABSOLUTE RATINGS: UHF oscillator, anode pulse modulated.

| Parameter: | F1 | Ef | Eb | Ib | eb | Pp | tk | T(seal) | Rk | Du | tp |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | MHz | V | V dc | mA dc | v | W | sec | °C | Ohms | --- | µs |
| Maximum: | 1,200 | 6.6 | 350 | 16.0 | 1,500 | 5.0 | --- | 175 | --- | 0.004 | 2.0 |
| Minimum: | --- | 6.0 | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| Test conditions: | --- | 6.3 | 180 | 12.0 | --- | --- | 300 | --- | 400 | --- | --- |

GENERAL:

Qualification - Not required.

Holding period (MIL-STD-1311): t = 168 hours.



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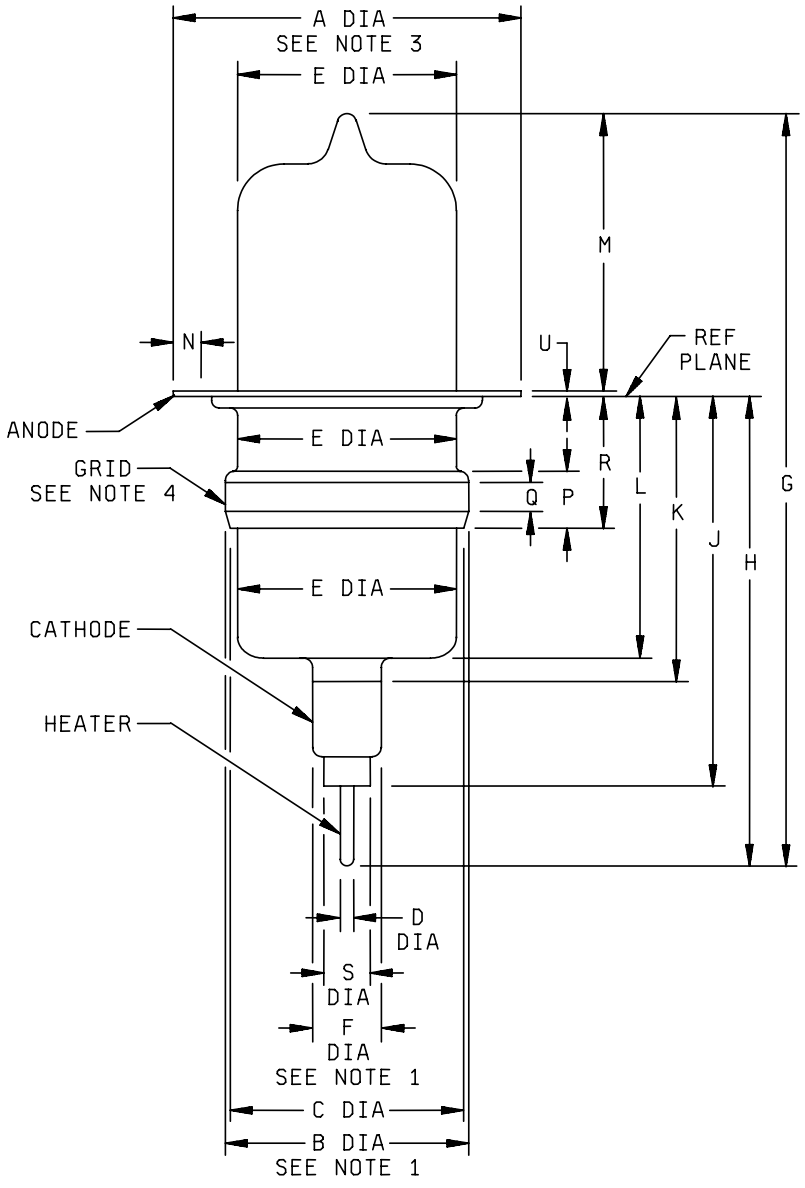


FIGURE 1. Outline drawing of electron tube type 2C36.



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| Letter | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Minimum | | Maximum | | |
| | Inches | mm | Inches | mm | |
| Conformance inspection, part 2 | | | | | |
| A | .995 | 25.27 | 1.005 | 25.53 | 1, 3 |
| B | .740 | 18.80 | .750 | 19.05 | 1, 3 |
| C | .695 | 17.65 | .725 | 18.42 | --- |
| D | .045 | 1.14 | .055 | 1.40 | 1, 2 |
| E | --- | --- | .660 | 16.76 | 1 |
| F | .197 | 5.00 | .207 | 5.26 | 1, 2 |
| G | --- | --- | 2.375 | 60.33 | --- |
| H | 1.405 | 35.69 | 1.475 | 37.47 | --- |
| J | 1.200 | 30.48 | 1.230 | 31.24 | --- |
| K | --- | --- | .900 | 22.86 | --- |
| L | --- | --- | .815 | 20.70 | --- |
| M | --- | --- | .887 | 22.53 | --- |
| N | .080 | 2.03 | --- | --- | 2 |
| P | .175 | 4.45 | .195 | 4.95 | --- |
| Q | .090 | 2.29 | --- | --- | 2 |
| R | .375 | 9.53 | .425 | 10.80 | --- |
| S | --- | --- | .190 | 4.83 | --- |
| U | .007 | 0.18 | .013 | 0.33 | --- |

NOTES:
1. Reference diameter "F" to be concentric with respective diameters as follows: Diameter "A" .040 (1.02 mm) FIR, diameter "B" .030 (0.76 mm) FIR, diameter "D" .030 (0.76 mm) FIR, diameter "E" .060 (1.52 mm) FIR. Full indicator reading includes out of roundness and tilt of components.
2. Areas designated by dimensions "D", "F", "N", and "Q" must be clean and free from glass.
3. Reference diameter "A" to be concentric with diameter "B" within .030 (0.76 mm) FIR.
4. This surface to be flat within .005 (0.13 mm).

FIGURE 1. Outline drawing of electron tube type 2C36 - Continued.



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TABLE I. Testing and inspection.

| MIL-STD-1311 method | Requirement or test | Conditions | Acceptance level (see note 4) | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| | Conformance inspection, part I | | | | | | |
| 1301 | Heater current | | 0.65 | If | 380 | 420 | mA |
| 1256 | Electrode current (1) (anode) | | 0.65 | Ib | 8.0 | 16.0 | mA dc |
| 1256 | Electrode current (2) (anode) | Ec = -28 V dc; Rk = 0 | 0.65 | Ib | --- | 10.0 | µA dc |
| 1306 | Transconductance | | 0.65 | Sm | 3,350 | 5,650 | µmhos |
| 1231 | Emission | (Eb = Ec)/Ik = 30 mA dc; Rk = 0 (see note 2) | 0.65 | Etd | --- | 10 | V dc |
| 1231 | Peak emission | eb = ec = 100 v; tp = 3.0 µs; prr = 500 | 0.65 | is | 800 | --- | mA |
| | Conformance inspection, part 2 | | | | | | |
| --- | Power output | F = 1,075 MHz ± 25 MHz; eb = 1,000 v; prr = 2,000; tp = 1.0 µs; cavity in accordance with 171-JAN | 6.5 | po | 125 | --- | w |
| 1031 | Vibration | No voltages | 6.5 | --- | --- | --- | --- |
| 1266 | Total grid current | | 6.5 | Ic | 0 | -1.0 | µA dc |
| 1211 | Insulation of electrodes | Ef = 6.3 V; Eg to all = 100 V dc | 6.5 | R | 25 | --- | MegΩ |
| 1316 | Amplification factor | | 6.5 | Mu | 15 | 35 | --- |
| 1331 | Direct-interelectrode capacitance | Test in cavity in accordance with 170-JAN | 6.5 | Cgp<br>Cgk<br>Cpk | 1.75<br>1.05<br>0.28 | 2.05<br>1.35<br>0.48 | pF<br>pF<br>pF |
| --- | Torque | Applied between anode and cathode without shock (see note 3) | 6.5 | --- | 10 | --- | lb-in. |
| --- | Seal | T = 175°C (min); t = 16 hours (min) (see note 1) | 6.5 | --- | --- | --- | --- |
| | Conformance inspection, part 3 | | | | | | |
| --- | Life-test provisions | Group C; power output t = 250 hours | --- | --- | --- | --- | --- |
| --- | Life-test end point | Power output | --- | po | 100 | --- | w |

See notes at top of next page.



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TABLE I. Testing and inspection - Continued.

NOTES:

1. This test shall consist of subjecting the tube to the temperature and minimum time specified, after which the tube shall be tested for heater current and the value shall be within 380 mA to 420 mA.
2. May be made at constant voltage as an alternate with Eb/-Ec = 10 V dc; Rk = 0; Is = 35 mA dc (min).
3. After subjection to the torque specified, the tube shall be tested for total grid current and the value shall not exceed -1.2 μA dc.
4. This specification sheet uses accept on zero defect sampling plan, in accordance with MIL-PRF-1, table III.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-1, this document references the following:

MIL-STD-1311
170-JAN
171-JAN

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-005)

Review activities:
Army - CR4, MI
Navy - AS, CG, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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